Stochastic process simulation and Euler-Maruyama scheme

Jan Rems

2025

Use the package Torch and tensor objects to store the processes for all the exercises.

1 Brownian Motion

Exercise 1.1. *Discretize interval* $[0,1]$ *into 500 points and generate one trajectory of Brownian motion. Try to do it using a for loop and also without it. Plot the processes and see if the results are the same. Repeat for a few realizations.*

Exercise 1.2. *Generate 1000 realizations of the exponential Brownian motion* $M_t = \exp(B_t - \frac{t}{2})$. *Estimate* $E(M_t)$ *for each* $t \in [0,1]$ *and comment. Plot 5 realization and* $E(M_t)$ *on the same graph.*

Exercise 1.3. *We define a running maximum of Brownian motion as* $M_t = \sup_{0\leq s\leq t} B_t$. *Show numerically that* $|B_t|$ *has the same distribution as* M_t *for each* $t \in [0,1]$.

2 Homogeneous Poisson Process

Exercise 2.1. *There are three common approaches for generating a homogeneous Poisson process. Generate 10000 realizations of* $N_t \sim HPP(10)$ *on* $[0,1]$ *using these three different approaches. Construct a test that will confirm that the approaches yield the same results.*

3 Euler-Maruyama scheme

Consider a Δ_t-distant time partition $\pi = \{0 = t_0 < t_1 < \cdots < t_N\}$ and an SDE

$$dX_t = b(X_t)dt + \sigma(X_t)dB_t, \quad X_0 = x_0.$$

Euler-Maruyama scheme for a numerical solution to the above SDE on partition π is given by a recursive formula $\hat{X}_0 = x_0$ and

$$\hat{X}_{i+1} = \hat{X}_i + b(\hat{X}_i)\Delta_t + \sigma(\hat{X}_i)\Delta B_i,$$

where $\Delta B_i = B_{t_{i+1}} - B_{t_i}$.

Exercise 3.1. *Using EM scheme, approximate the solution of the SDE*

$$dX_t = \theta(\mu - X_t)dt + \sigma dB_t$$

where $X_0 = 0, \mu = 0.5, \theta = 2, \sigma = 0.2$. *Estimate* $P(X_5 \leq \mu)$. *Show graphically that* $E[X_t]$ *converges towards* μ. *Compare the obtained result to the explicit solution*

$$X_t = X_0 e^{-\theta t} + \mu(1 - e^{-\theta t}) + \sigma \int_0^t e^{-\theta(t-s)} dB_s.$$

Exercise 3.2. *For SDE*

$$dX_t = 0.3X_t dt + 0.2X_t dB_t, \quad X_0 = 1,$$

use 2^9 *time-points to define a discrete version of the exact solution given by:*

$$X_t = \exp\left((0.3 - \frac{0.2^2}{2})t + 0.2B_t\right).$$

Then use EM scheme for $\Delta_t^j = \Delta_t 2^{-j}$ *for* $j = 0, \ldots, 5$ *and estimate the rates of convergence*

- $\sup_{0\leq t\leq T} E(|X_t - \hat{X}_t|)$,
- $E(\sup_{0\leq t\leq T} |X_t - \hat{X}_t|)$,
- $|E(X_T) - E(\hat{X}_T)|$.

Draw a graph for the estimated rates and compare it with theoretical ones. Use 100000 sample paths for estimation.

4 Black-Scholes Model

Exercise 4.1. *Assume the Black-Scholes model with interest rate* $r = 0.05$, *average return rate* $\mu = 0.1$ *and volatility* $\sigma = 0.2$. *Denote with* S_t *a stock price where* $S_0 = 100$. *Assume a European call option with maturity* $T = 1$ *and a strike price* $K = 105$. *Using Monte-Carlo, approximate the discounted stock price* $V_0 = E_Q[(S_T - K)_+]$ *where* Q *is a risk-neutral measure. Compare the result with the one obtained through the BS formula.*